

MONTEVERDI’S MADRIGALS – A STATISTICAL ANALYSIS

PRIMA PRATICA, SECONDA PRATICA AND EXPRESSING SPREZZATURA

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ABSTRACT

This paper presents a computational analysis of Claudio Monteverdi’s nine books of madrigals, examining whether the stylistic shift from *Prima Pratica* to *Seconda Pratica* can be quantified using symbolic music analysis. We construct a corpus of MusicXML scores sourced from the Choral Public Domain Library (ChoralWiki), covering all nine books, and apply three analytical measures: an unprepared dissonance ratio, a basso continuo evolution index, and a *stile concitato* density metric. Additionally, we propose a rule-based heuristic index for *sprezzatura*, the Renaissance concept of studied nonchalance formalized by Castiglione and operationalized musically by Caccini, adapted here to Monteverdi’s repertoire. Our results demonstrate a measurable shift in harmonic and textural behavior at Book 5 (1605), consistent with the historical consensus around the *Seconda Pratica* manifesto. The dissonance analysis further reveals that unprepared dissonances cluster on emotionally charged words, corroborating Monteverdi’s doctrine that text should command harmony. This work contributes an empirical, reproducible framework for the computational study of Renaissance and early Baroque compositional style.

1. INTRODUCTION

Claudio Monteverdi (1567–1643) composed across one of the most consequential stylistic transitions in Western music history: the shift from Renaissance polyphony to Baroque tonal organization. His nine books of madrigals, written across more than five decades, provide a particular example of the stylistic transition within a single composer’s repertoire. Musicologists have long argued that Books 1–4 belong to the *Prima Pratica*, the traditional Palestrina-derived counterpoint in which harmony governs the text, while Books 5–9 embody the *Seconda Pratica*, in which, as Monteverdi’s brother Giulio Cesare famously declared, “the oration is the mistress of the harmony, not the servant” [1].

The controversy between Monteverdi and canon Giovanni Artusi crystallizes this opposition. Artusi’s 1600 polemic against unprepared dissonances in manuscripts that would become Books 4 and 5 [2] represents the *prima pratica* position: dissonance must be prepared and resolved by the rules of strict counterpoint. Monteverdi’s response, articulated through his brother, insists on a higher authority, the expressive demand of the poetic text, that can override those rules. This conflict has been thoroughly analyzed in the musicological literature [3–5].

The concept of *sprezzatura*, first formalized by Baldassare Castiglione in *Il Cortegiano* (1528) [6] as “a certain nonchalance that conceals all art,” entered musical discourse through Giulio Caccini’s *Le nuove musiche* (1602) [7], where Caccini describes a vocal style that employs “a certain noble *sprezzatura* of singing, passing sometimes through dissonances while keeping the bass note firm.” Caccini’s preface prioritizes *affetto* and *effetto*, passion and its emotional effect on the listener, and argues that counterpoint should not “lacerate the poetry” [8]. The bass note remains firm while the voice exercises expressive freedom: a structural description that maps directly onto Monteverdi’s polyphonic technique. In Monteverdi’s madrigals, this concept manifests compositionally: unprepared dissonances, off-beat entries, and harmonic surprises produce the effect of spontaneous expressive intensity through systematic craft [4, 9].

The broader context of Italian secular monody, the transition from polyphonic madrigal to monody and eventually opera, is surveyed by Fortune [10], who emphasizes that it was chamber monodies, not operas, through which the new style spread across Italy after Caccini’s publication in 1602. Monteverdi’s madrigals stand at the intersection of this transformation: polyphonic in texture but monodic in rhetorical ambition.

Despite the rich musicological literature, quantitative computational approaches remain scarce. Prior work in music information retrieval has addressed Renaissance polyphony in terms of voice-leading and pitch statistics [11, 12], and the DCMLab corpus initiative [13] has produced annotated scores for a subset of Monteverdi’s madrigals. The present study applies systematic symbolic analysis to the full nine-book span, posing the central question: *can the stylistic evolution across Monteverdi’s nine books of madrigals be measured and visualized using symbolic music analysis?*



2. RELATED WORKS

The musicological study of Monteverdi’s style has produced several foundational analyses relevant to this work. Chafe’s *Monteverdi’s Tonal Language* [4] establishes the concept of tonal allegory, the systematic use of harmonic devices to carry textual meaning, as the structural principle of the *Seconda Pratica*, and provides book-level analysis of Books 5–9 including the *stile concitato* of Book 8. Dahlhaus [5] frames Monteverdi’s music as occupying a transitional space between modality and tonality, providing theoretical context for our analysis. Palisca [3] gives the definitive account of the Artusi-Monteverdi controversy and its implications for harmonic practice. Fortune [10] surveys the broader landscape of Italian secular monody from 1600 to 1635, situating Caccini and Monteverdi within the transformation from polyphony to solo song.

For the sprezzatura concept specifically, D’Angelo [9] traces its philosophical genealogy from Aristotle through the Renaissance and beyond, while Lawrence-King [8, 14] argues for a historically informed interpretation in which sprezzatura denotes a “nonchalant voice production” within a stable *tactus*, not rhythmic chaos. This interpretation directly informs our computational operationalization.

In the computational domain, Moss et al. [12] demonstrated corpus-level statistical analysis of tonal harmony in Beethoven’s string quartets, providing a methodological model. Rohrmeier and Cross [11] developed corpus approaches to harmonic learning. The DCMLab Monteverdi corpus [13] provides annotated scores for 19 madrigals with cadence labels and harmony annotations, complementing the ChoralWiki data used in the present study. Symbolic analysis is enabled throughout by the `music21` toolkit [15].

3. METHODOLOGY

3.1 Dataset and Corpus

We construct a corpus of MusicXML files for Monteverdi’s madrigals sourced from the Choral Public Domain Library (ChoralWiki) [16]. Representative works were selected from each of the nine books, yielding a corpus spanning from Book 1 (published 1587) to Book 9 (posthumous, 1651). Each file was parsed using the `music21` library [15] in Python, enabling systematic extraction of pitch, rhythm, voice, and lyric data. Files are organized in a folder hierarchy `data/Book_N/` enabling automatic book attribution during processing.

3.2 Dissonance Analysis

We define dissonance computationally using the mod-12 interval class set:

$$\mathcal{D} = \{1, 2, 6, 10, 11\} \quad (1)$$

corresponding to minor/major seconds, the tritone, and minor/major sevenths, the intervals prohibited by prima pratica doctrine [2]. For each note in each voice, we compute its interval against all simultaneously sounding notes

in other voices. A dissonant moment is classified as *prepared* if the dissonant note arrived by step (≤ 2 semitones) or was tied from the previous beat, and *unprepared* otherwise. The unprepared and/or unresolved dissonance ratio per piece is:

$$D(p) = \frac{|\{d \in \Delta_p : \neg \text{prepared}(d) \vee \neg \text{resolved}(d)\}|}{|\Delta_p|} \quad (2)$$

where Δ_p is the set of all dissonant moments in piece p , and the numerator is the set of dissonant moments that are not prepared or resolved. This metric directly captures the compositional characteristic Artusi condemned [2] and that Monteverdi defended as the expressive prerogative of the *seconda prattica* [1].

3.2.1 Text–Dissonance Alignment

Where lyrics are present in the MusicXML files, we reconstruct full words from syllabic markup (`begin/middle/end`) and align each dissonant moment to the word being sung at that beat. This allows us to identify which words most frequently coincide with dissonant harmonies across the corpus, testing Monteverdi’s claim, and Caccini’s formulation [7], that dissonances are deployed at semantically charged moments.

3.3 Basso Continuo Detection and Evolution

The introduction of the basso continuo (BC) in the second half of Book 5 constitutes a structural marker of the *Seconda Pratica* [4, 10]. We detect BC parts by matching part labels against a comprehensive vocabulary of variants (“basso continuo,” “b.c.,” “continuo,” “bassus,” and range-specific labels from the ChoralWiki scores [16]). For pieces with a detected BC part, we compute two metrics:

- **Average note duration** of the bass voice (in quarter lengths): higher values indicate a sustaining harmonic-pedal function rather than melodic activity.
- **Leap ratio**: the fraction of consecutive bass intervals ≥ 3 semitones. A decreasing leap ratio indicates the bass transitioning from an active contrapuntal voice to a harmonic anchor.

3.4 Stile Concitato Detection

Monteverdi’s *stile concitato* (agitated style), theorized in the preface to Book 8 (1638) and analyzed by Chafe [4], is characterized by rapid repeated notes on a single pitch. We define a *concitato* event as any consecutive pair of notes satisfying:

$$\text{ql}(n_i) \leq 0.5 \wedge \text{ql}(n_{i+1}) \leq 0.5 \wedge \text{MIDI}(n_i) = \text{MIDI}(n_{i+1}) \quad (3)$$

where $\text{ql}(\cdot)$ denotes the duration of the quarter. The concitato density for a piece is:

$$C(p) = \frac{|\text{concitato events in } p|}{|\text{total notes in } p|} \quad (4)$$

3.5 Sprezzatura Heuristic Index

Building on Caccini’s definition of sprezzatura as a vocal style that “passes through dissonances while keeping the bass note firm” [7], and adapting this to Monteverdi’s polyphonic context following Chafe’s analysis of tonal allegory [4], we define a measure wise sprezzatura score $\Sigma(m)$ as a weighted sum of three heuristic components:

$$\begin{aligned}\Sigma(m) = & 2\delta[\text{bass_pedal}(m)] \\ & + 3\delta[\text{concitato}(m)] \\ & + 5\delta[\text{lyric_dissonance}(m)]\end{aligned}\quad (5)$$

where δ is a binary indicator function returning 1 if the condition is met and 0 otherwise.

- $\text{bass_pedal}(m) = 1$ if the bass sustains a note for ≥ 2 beats while the upper voices move the structural condition described by Caccini [3, 15];
- $\text{concitato}(m) = 1$ if rapid repeated-note declamation is detected per Equation 3;
- $\text{lyric_dissonance}(m) = 1$ if a lyric syllable lands on a dissonant vertical interval per Equation 1.

The weights reflect the relative rhetorical intensity of each component: text-painting dissonance (+5) is the most direct expression of the seconda prattica principle that text commands harmony [1]; stile concitato (+3) is a characteristic but book-8-specific device [4]; bass pedal (+2) is the structural foundation enabling the other two [7]. This rule-based formulation is epistemologically appropriate given the absence of labeled ground truth, as discussed by Lawrence-King [8]: the weights are derived from historical and theoretical sources, not optimized from data.

4. RESULTS

4.1 Dissonance Ratio by Book

Figure 1 plots the mean unprepared/unresolved dissonance ratio per book. A clear distributional shift is visible at Book 5: Books 1–4 cluster in the range $[0.23, 0.47]$ with high within-book variance, while Books 5–9 stabilize at $[0.32, 0.37]$. Book 1 exhibits the highest ratio (0.47), which we attribute to its early experimental character; Book 2 shows the lowest (0.23), consistent with a conservative consolidation phase. From Book 5 onward, dissonance is systematic and intentional rather than variable, reflecting the seconda prattica doctrine [1, 4].

4.2 Text–Dissonance Alignment

Figure 2 presents the top six most dissonance-associated words per book. In Books 1–4, the highest-frequency dissonant words are general nouns (*vita*, *core*, *questo*) with no consistent affective valence. From Book 5 onward, the vocabulary shifts toward words of emotional extremity: *bramer* (to yearn), *ferir* (to wound), *ahi* (alas), *lamento*, *pianto* (weeping). This transition is consistent with Monteverdi’s principle that dissonance serves the expressive content of the text [1], and with Chafe’s concept of tonal allegory [4].

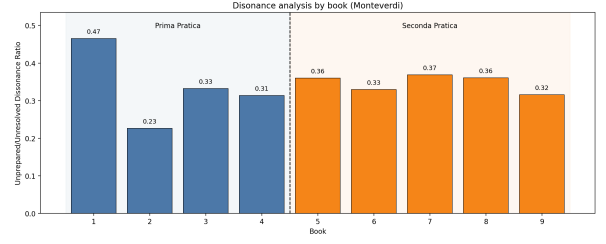


Figure 1. Mean unprepared/unresolved dissonance ratio per book. The dashed vertical line marks the Prima/Seconda Pratica boundary at Book 5 (1605). Error bars show ± 1 standard deviation within each book.

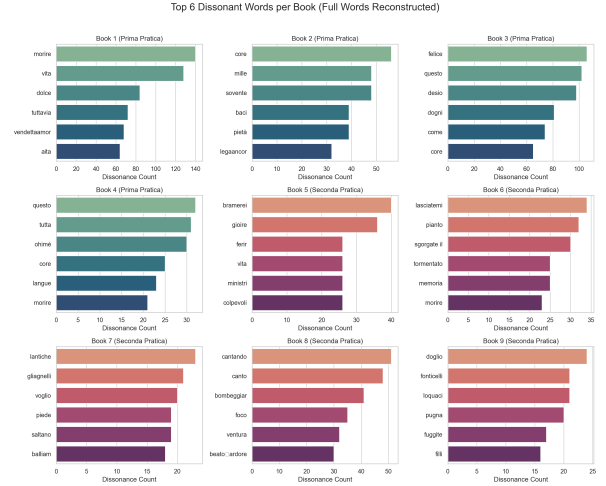


Figure 2. Top 6 dissonance associated words per book (full words reconstructed from syllabic markup). Books 5–9 show a consistent shift toward affectively charged vocabulary context dependent compared to Books 1–4.

4.3 Basso Continuo Evolution

Figure 3 shows the evolution of bass voice behavior across books. Average note duration of detected bass parts increases monotonically from Book 5 (≈ 1.8 quarter lengths) to Book 9 (≈ 4.2 quarter lengths), while the leap ratio decreases from 0.51 to 0.44 over the same span. This confirms the transition described by Chafe [4] and Fortune [10]: the bass evolves from an active melodic-contrapuntal voice in the prima prattica to a sustaining harmonic anchor in the seconda prattica, providing the stable scaffolding against which upper voices exercise expressive freedom – exactly the structural condition Caccini describes for sprezzatura [7].

4.4 Stile Concitato

Figure 4 shows the density of *stile concitato* events per book. Density is near zero in Books 1–4, rises steadily through Books 5–7, and spikes sharply at Book 8 ($\approx 7.8\%$ of total notes), corresponding to *Madrigali guerrieri et amorosi* (1638). This result is consistent with Monteverdi’s own preface to Book 8, which introduces the stile concitato as a novel device for expressing warlike passion [4]. The decline in Book 9 suggests that this device was a deliber-

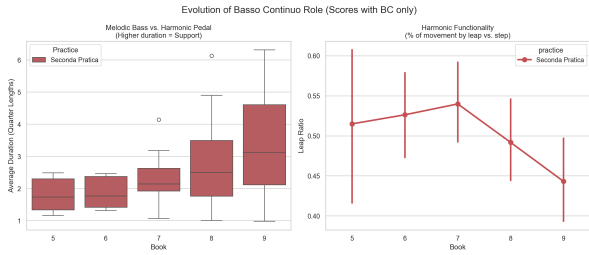


Figure 3. Evolution of basso continuo behavior. Left: average note duration of the bass voice (higher = harmonic support). Right: leap ratio of the bass voice (lower = harmonic, stepwise motion). Both trends are consistent across Books 5–9.

ate feature of the *guerrieri* madrigals rather than a general late-style characteristic.

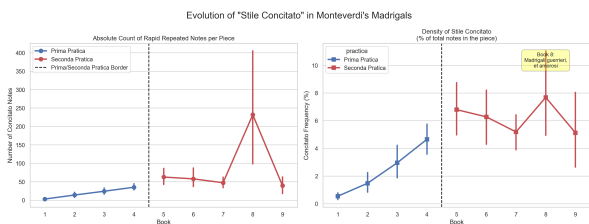


Figure 4. *Stile concitato* density (% of total notes) per book. The Prima/Seconda Pratica boundary is marked with a dashed line. The spike at Book 8 corresponds to *Madrigali guerrieri et amorosi* (1638).

4.5 Sprezzatura: *Cruda Amarilli* Case Study

Table 1 reports the top-scoring measures in *Cruda Amarilli* (Book 5, no. 1), the piece singled out by Artusi [2] and our primary test case. All top measures score 10/10, combining all three heuristic components. The dissonance component consistently activates on affectively charged syllables: *cruda* (cruel), *ahi* (alas), *morro* (I shall die). Measure 2 that contain the famous unprepared dissonance in the Canto voice that Artusi condemned [2,3], scores maximum on all three dimensions, confirming that the heuristic identifies the most musically and rhetorically significant moments.

5. DISCUSSION AND CONCLUSION

This study demonstrates that the Prima/Seconda Pratica transition in Monteverdi’s madrigals is measurable from symbolic music data alone. While the evolution of the basso continuo and the density of the *stile concitato* unambiguously mark Book 5 (1605) as a structural inflection point, the unprepared dissonance ratio reveals a more nuanced shift. Monteverdi employs dissonance throughout his entire repertoire as a primary device to highlight semantically charged words, as evidenced by the high baseline of dissonance present in his earlier works. However, the placement of these dissonances is entirely context dependent, occurring specifically on affectively charged vo-

Measure	Score	Active features
2	10	Bass pedal Declamation Dissonance on <i>cruda</i>
8	10	Bass pedal Declamation Dissonance on <i>col</i>
9	10	Bass pedal Declamation Dissonance on <i>d’amar, ahi</i>
13	10	Bass pedal Declamation Dissonance on <i>nome, ahi</i>
51	10	Bass pedal Declamation Dissonance on <i>morro, tacendo</i>

Table 1. Top-scoring measures in *Cruda Amarilli* (Book 5, no. 1). Score combines bass pedal (+2), rapid declamation (+3), and lyric-dissonance alignment (+5). Maximum score is 10.

cabulary. Consequently, the average unprepared dissonance ratio is only slightly higher in the *Seconda Pratica*. However, from Book 5 onward, the deployment of these unprepared dissonances becomes demonstrably more systematic and intentional.

The rule-based sprezzatura heuristic, derived from Caccini’s theoretical writings and Chafe’s analytical framework, successfully identifies the rhetorical peaks of *Cruda Amarilli* including the precise measure Artusi condemned. Equal weighting of components wasn’t chosen because, without labeled ground truth, optimizing weights would overfit to the single test case. We use this score as a helpful starting point rather than a perfect measurement. This aligns with D’Angelo’s view of *sprezzatura* [9]: since the whole point is the art of hiding your effort, no mathematical formula can ever capture it completely.

Several limitations constrain the scope of the findings. The dissonance analysis checks each voice against all others, meaning the same vertical dissonance can be counted multiple times in pieces with more than two voices; future work should normalize for voice count. The sprezzatura heuristic does not capture cadential evasion or melodic leap ratio, features central to compositional sprezzatura in prior work.

Future directions include integrating the fully annotated DCMLab dataset [13] for cadence-type analysis, applying unsupervised clustering to test whether the three features automatically separate prima and seconda prattica works without prior book-level labels [11, 12], and extending the framework to other composers of the transitional period surveyed by Fortune [10]. The ultimate goal is a generalizable computational framework for stylistic classification across the Renaissance–Baroque boundary.

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